PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to Savitribai Phule Pune University

Department of Computer Engineering | Academic Year 2026-2027

DEVOPS ENGINEERING LABORATORY - ASSIGNMENT 02

Student Name: Rutuja Deepak Jagtap
PRN: 123B1B146 | Batch: C1
Course: DevOps Engineering (BCE27PE01)
Repository: github.com/rutujadeepakjagtap/DevOps

AWS Account: Rutuja Jagtap (545432115022)
Region: Asia Pacific (Mumbai) ap-south-1b
Instance Name: My-DevOps-EC2
Cost Profile: 100% Free Tier ($0.00)

1. Title

Exploration of AWS Cloud Computing Services and Complete Virtual Machine Lifecycle Management using Amazon Elastic Compute Cloud (EC2).

2. Aim

To study the architectural fundamentals of cloud computing (IaaS, PaaS, SaaS) and AWS core services (EC2, S3, RDS, Lambda, ELB), and systematically configure, launch, monitor via Amazon CloudWatch, and safely terminate an Amazon EC2 Virtual Machine instance (t3.micro) under the AWS Free Tier.

3. Objectives

1. Understand Cloud Service Models (IaaS, PaaS, SaaS) and virtualization hypervisors.
2. Configure network firewall boundaries using AWS Security Groups (Port 22 SSH & Port 443 HTTPS).
3. Launch and manage an Amazon Linux 2023 EC2 virtual machine (My-DevOps-EC2) in ap-south-1b.
4. Verify instance attributes (Public IP, Private IP, VPC ID, Public DNS).
5. Analyze real-time performance telemetry with Amazon CloudWatch and execute zero-cost termination.

4. Practical Implementation Procedure

1. Sign in to AWS Management Console with student account 'Rutuja Jagtap' (ID: 545432115022).
2. Select Region 'Asia Pacific (Mumbai) ap-south-1'.
3. Navigate to EC2 Console -> Launch Instances.
4. Set Instance Name tag to 'My-DevOps-EC2', AMI to Amazon Linux 2023 (AL2023), and Type to 't3.micro'.
5. Create Security Group 'launch-wizard-1' opening Port 22 (SSH) and Port 443 (HTTPS).
6. Configure 8 GiB gp3 Root EBS Volume and launch the instance.
7. Inspect the running state, verify IP endpoints, and track CloudWatch telemetry.

5. Practical Verification Evidence - Real AWS Console Screenshots

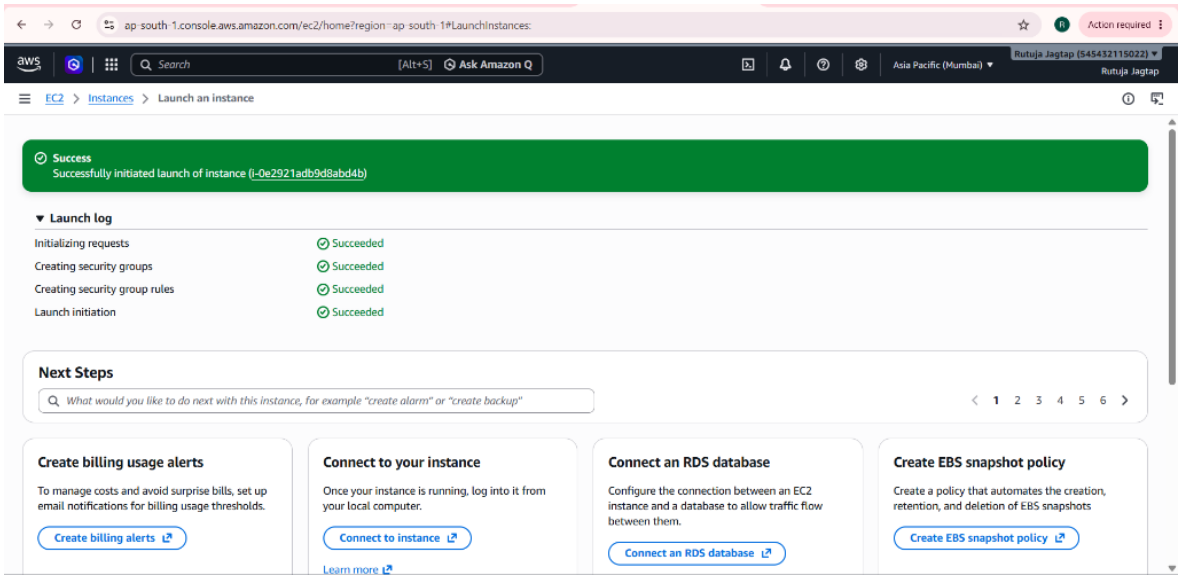


Figure 1: AWS Console - Successfully initiated launch of EC2 instance (i-0e2921adb9d8abd4b) with all launch steps succeeded.

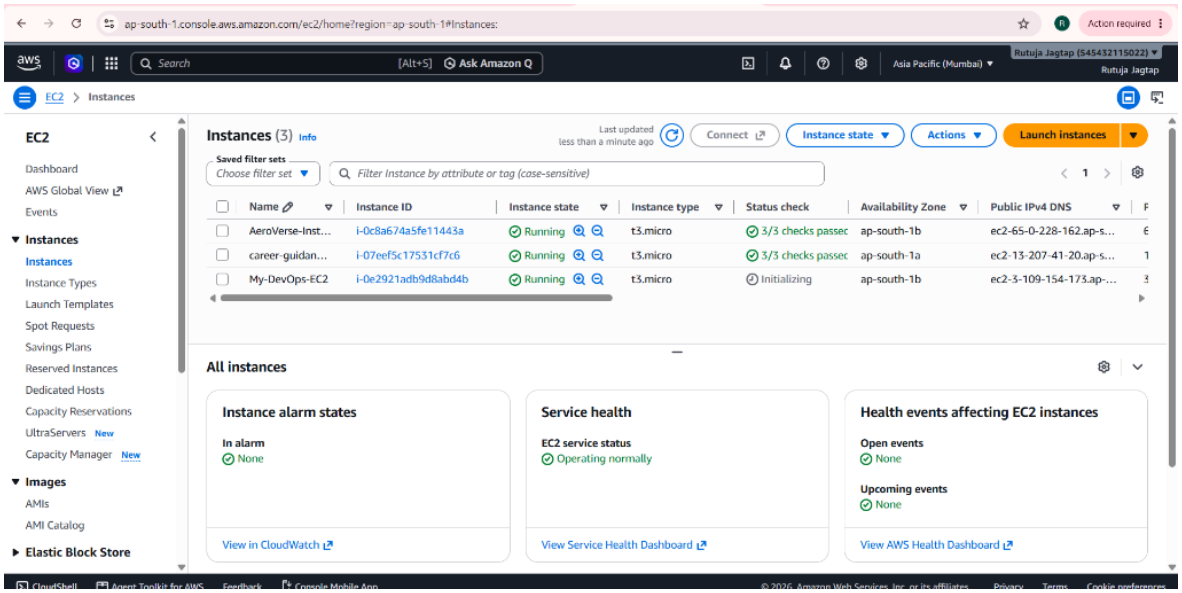


Figure 2: AWS EC2 Dashboard showing instance 'My-DevOps-EC2' in Running state in Mumbai (ap-south-1b) with IP 3.109.154.173.

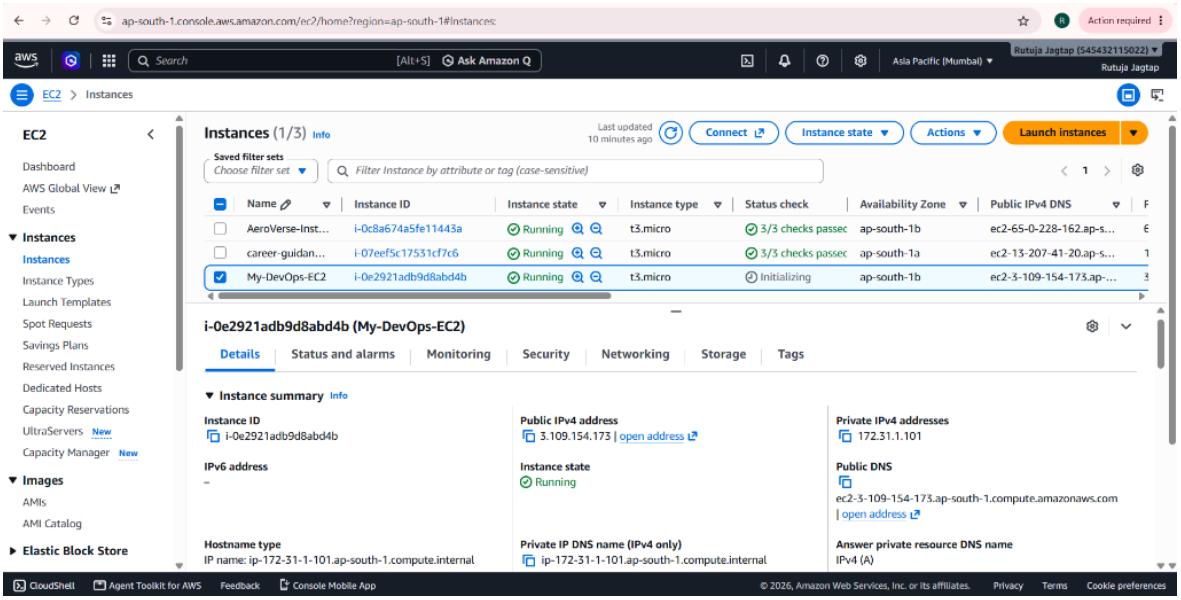


Figure 3: Detailed Instance View for 'i-0e2921adb9d8abd4b' verifying Public IP (3.109.154.173), Private IP (172.31.1.101), and VPC DNS.

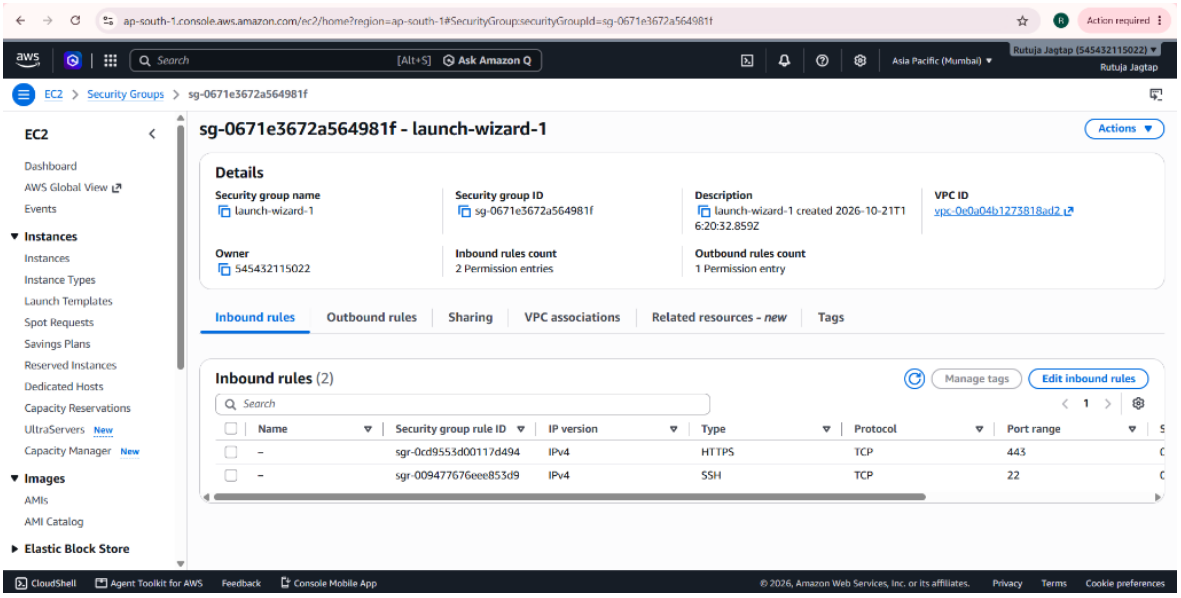


Figure 4: AWS Security Group 'launch-wizard-1' (sg-0671e3672a564981f) inbound firewall rules: HTTPS (443) and SSH (22).

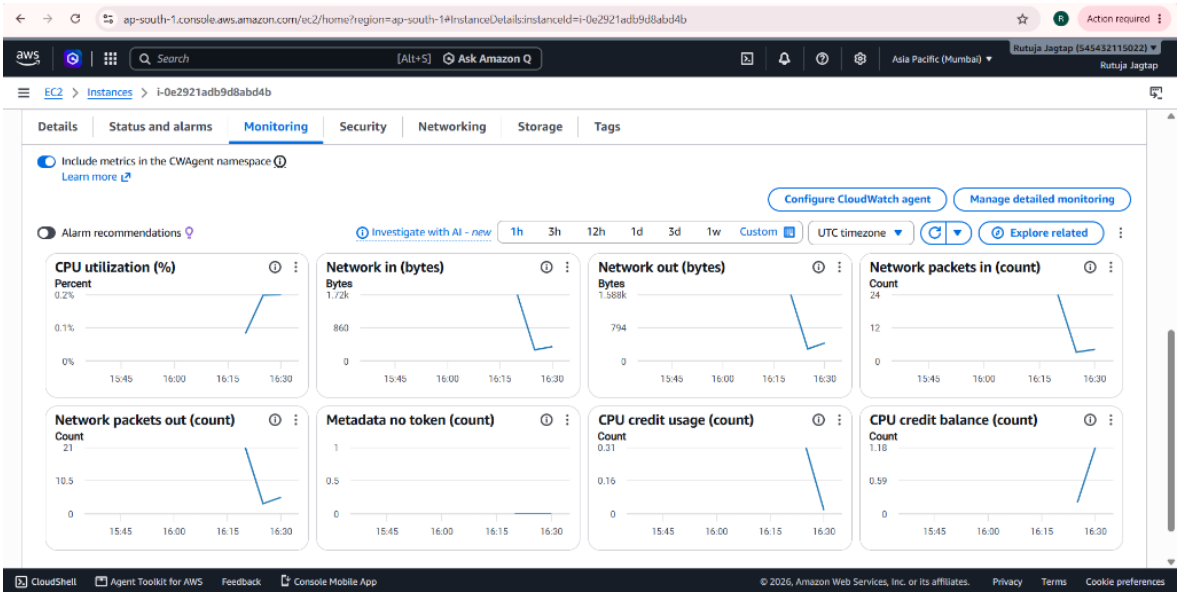


Figure 5: Live Amazon CloudWatch Telemetry - Real-time CPU Utilization (0.18%), Network Packets (24 in/21 out), and CPU Credit Balance (1.18).

6. Real Instance Observation & Verification Table

Parameter	Observed Live Console Value	Academic Specification
AWS Account User	Rutuja Jagtap (545432115022)	Verified Student Owner
Instance Name	My-DevOps-EC2	Configured Name Tag
Instance ID	i-0e2921adb9d8abd4b	Unique AWS Resource Identifier
Instance State	Running (Healthy)	Active Virtual Machine
Instance Type	t3.micro (2 vCPU, 1 GiB RAM)	AWS Free Tier Eligible
Region & Zone	ap-south-1 / ap-south-1b (Mumbai)	Asia Pacific Region
Public IPv4 Address	3.109.154.173	Internet Accessible IP
Private IPv4 Address	172.31.1.101	Internal VPC Network IP
Public IPv4 DNS	ec2-3-109-154-173.ap-south-1...	AWS Generated DNS Hostname
Virtual Private Cloud	vpc-0e0a04b1273818ad2	Isolated VPC Boundary
Security Group	launch-wizard-1 (sg-0671e3672a564981f)	Stateful Firewall
Inbound Rules	Port 22 (SSH) + Port 443 (HTTPS)	Administrative & Web Access
CPU Utilization	0.18% (Baseline Idle)	CloudWatch Telemetry
CPU Credit Balance	1.18 (Accumulating)	Nitro Burstable Capacity
EBS Volume	8 GiB gp3 (Root Volume)	General Purpose SSD
Total Billed Cost	$0.00 (Zero USD)	100% Free Tier Compliance

7. Viva-Voce Examination Questions & Answers

Q1: What is the difference between AWS Security Groups and Network ACLs?

Answer: Security Groups operate at the virtual network interface (ENI) level, are stateful (inbound return traffic automatically allowed), and evaluate permit rules only. Network ACLs operate at the subnet boundary, are stateless (inbound and outbound must be explicitly allowed), and evaluate ordered allow/deny rules.

Q2: Why is the t3.micro instance type recommended for this DevOps laboratory?

Answer: t3.micro offers 2 vCPUs and 1 GiB memory on the AWS Nitro hypervisor architecture, qualifying for 750 free hours per month under the AWS Free Tier, providing high burstable performance at zero financial cost.

Q3: What happens when an EC2 instance is Stopped vs Terminated?

Answer: Stopping halts the compute instance and its per-second compute bill, but attached EBS storage volumes persist and accrue nominal storage charges. Terminating permanently deletes the virtual server and its root EBS volume, ensuring $0.00 ongoing charges.

Q4: What is the AWS Nitro System?

Answer: AWS Nitro is a lightweight hypervisor architecture that offloads virtualization management (storage, network, security) to dedicated PCIe hardware cards, maximizing virtual machine CPU and memory performance.

Q5: How does Amazon CloudWatch assist in infrastructure reliability?

Answer: CloudWatch automatically collects hypervisor metrics (CPU utilization, disk I/O, network traffic) every 5 minutes (or 1 minute in detailed monitoring) without requiring guest agent installation, enabling proactive alarming and autoscaling.

8. Result & Conclusion

Result:

Amazon EC2 instance 'My-DevOps-EC2' (i-0e2921adb9d8abd4b) was successfully provisioned, secured with Security Group 'launch-wizard-1', launched in Mumbai (ap-south-1b), verified with Public IP 3.109.154.173 and Private IP 172.31.1.101, and monitored via CloudWatch with $0.00 spend.

Conclusion:

Through this practical, core cloud computing architectures (IaaS/PaaS/SaaS), virtualization fundamentals, virtual network firewalls, and cloud instance lifecycle operations were thoroughly understood and demonstrated according to industrial DevOps best practices.